EMG4_story_fairness_data

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packages

```
library(readxl)
library(tidyverse)
library(ggthemes)
```

Fairness List 1

```
fairness1_filtered <- read_csv(paste0(getwd(),"/Qualtrics/EMG4_Vragenlijst_1.1_final.csv"),
                               show_col_types = FALSE) %>%
  slice(-c(1, 2)) %>%
  select(as.character(0:64))%>%
  mutate_at('0', as.factor) %>%
  mutate_at(as.character(1:64), as.numeric) %>%
  rename(sjCode = '0')

head(fairness1_filtered)
```

```
## # A tibble: 6 x 65
##   sjCode   `1`   `2`   `3`   `4`   `5`   `6`   `7`   `8`   `9`  `10`  `11`  `12`
##   <fct>  <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 8811       6     7     6     1     1     6     2     5     2     5     2     3
## 2 6012       7     6     6     2     2     6     2     6     4     4     2     2
## 3 5066       6     7     6     1     2     7     2     5     2     4     2     3
## 4 2009       7     7     7     1     1     7     1     7     1     6     1     1
## 5 7030       7     6     7     1     1     6     1     7     1     5     1     1
## 6 2528       4     4     4     2     2     4     3     5     4     4     3     4
## # i 52 more variables: `13` <dbl>, `14` <dbl>, `15` <dbl>, `16` <dbl>,
## #   `17` <dbl>, `18` <dbl>, `19` <dbl>, `20` <dbl>, `21` <dbl>, `22` <dbl>,
## #   `23` <dbl>, `24` <dbl>, `25` <dbl>, `26` <dbl>, `27` <dbl>, `28` <dbl>,
## #   `29` <dbl>, `30` <dbl>, `31` <dbl>, `32` <dbl>, `33` <dbl>, `34` <dbl>,
## #   `35` <dbl>, `36` <dbl>, `37` <dbl>, `38` <dbl>, `39` <dbl>, `40` <dbl>,
## #   `41` <dbl>, `42` <dbl>, `43` <dbl>, `44` <dbl>, `45` <dbl>, `46` <dbl>,
## #   `47` <dbl>, `48` <dbl>, `49` <dbl>, `50` <dbl>, `51` <dbl>, `52` <dbl>, ...
```

```r
fairness1_centered <- fairness1_filtered %>%
  mutate(across(c(2:65), ~ case_when(
    . == 1 ~ 3,
    . == 2 ~ 2,
    . == 3 ~ 1,
    . == 4 ~ 0,
    . == 5 ~ -1,
    . == 6 ~  -2,
    . == 7 ~ -3)))
head(fairness1_centered)
```

```
## # A tibble: 6 x 65
##   sjCode   `1`   `2`   `3`   `4`   `5`   `6`   `7`   `8`   `9`  `10`  `11`  `12`
##   <fct>  <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 8811      -2    -3    -2     3     3    -2     2    -1     2    -1     2     1
## 2 6012      -3    -2    -2     2     2    -2     2    -2     0     0     2     2
## 3 5066      -2    -3    -2     3     2    -3     2    -1     2     0     2     1
## 4 2009      -3    -3    -3     3     3    -3     3    -3     3    -2     3     3
## 5 7030      -3    -2    -3     3     3    -2     3    -3     3    -1     3     3
## 6 2528       0     0     0     2     2     0     1    -1     0     0     1     0
## # i 52 more variables: `13` <dbl>, `14` <dbl>, `15` <dbl>, `16` <dbl>,
## #   `17` <dbl>, `18` <dbl>, `19` <dbl>, `20` <dbl>, `21` <dbl>, `22` <dbl>,
## #   `23` <dbl>, `24` <dbl>, `25` <dbl>, `26` <dbl>, `27` <dbl>, `28` <dbl>,
## #   `29` <dbl>, `30` <dbl>, `31` <dbl>, `32` <dbl>, `33` <dbl>, `34` <dbl>,
## #   `35` <dbl>, `36` <dbl>, `37` <dbl>, `38` <dbl>, `39` <dbl>, `40` <dbl>,
## #   `41` <dbl>, `42` <dbl>, `43` <dbl>, `44` <dbl>, `45` <dbl>, `46` <dbl>,
## #   `47` <dbl>, `48` <dbl>, `49` <dbl>, `50` <dbl>, `51` <dbl>, `52` <dbl>, ...
```

```r
fairness1_centered_pivot <- fairness1_centered %>%
  pivot_longer(!sjCode, names_to = "Trial", values_to = "F_rating") %>%
  arrange(sjCode) %>%
  mutate_at("Trial", as.numeric)
head(fairness1_centered_pivot)
```

```
## # A tibble: 6 x 3
##   sjCode Trial F_rating
##   <fct>  <dbl>    <dbl>
## 1 1258       1       -2
## 2 1258       2       -1
## 3 1258       3       -2
## 4 1258       4        2
## 5 1258       5        2
## 6 1258       6       -3
```

```r
fairness1_filtered_pivot <- fairness1_filtered %>%
  pivot_longer(!sjCode, names_to = "Trial", values_to = "F_filt_rating") %>%
  arrange(sjCode) %>%
  mutate_at("Trial", as.numeric)
head(fairness1_filtered_pivot)
```

```
## # A tibble: 6 x 3
##   sjCode Trial F_filt_rating
```

```
##    <fct>  <dbl>          <dbl>
## 1 1258       1              6
## 2 1258       2              5
## 3 1258       3              6
## 4 1258       4              2
## 5 1258       5              2
## 6 1258       6              7
```

```r
table(fairness1_filtered_pivot$F_filt_rating)
```

```
##
##   1   2   3   4   5   6   7
## 178 164 105 113 150 153  97
```

```r
table(fairness1_centered_pivot$F_rating)
```

```
##
##  -3  -2  -1   0   1   2   3
##  97 153 150 113 105 164 178
```

save the questionnaire data to new csv file

```r
write_csv(fairness1_centered_pivot, file = paste0(getwd(),"/Qualtrics/fairness_1_EMG4_final.csv"))
```

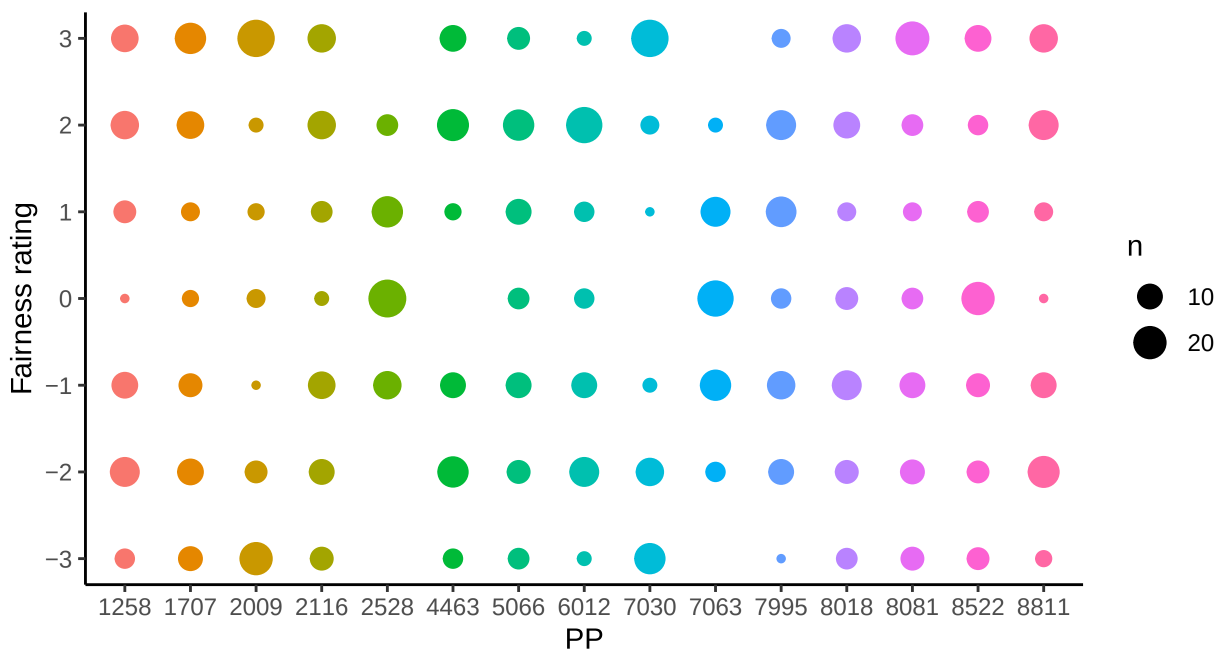

Figure 1: Variation in fairness ratings within participants (List 1)

Fairness List 2

```
fairness2_filtered <- read_csv(paste0(getwd(),"/Qualtrics/EMG4_Vragenlijst_1.2_final.csv"), show_col_ty
  slice(-c(1, 2)) %>%
  select(as.character(0:64))%>%
  mutate_at('0', as.factor) %>%
  mutate_at(as.character(1:64), as.numeric) %>%
  rename(sjCode = '0')

head(fairness2_filtered)
```

```
## # A tibble: 6 x 65
##   sjCode   `1`   `2`   `3`   `4`   `5`   `6`   `7`   `8`   `9`  `10`  `11`  `12`
##   <fct>  <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 7735       2     1     6     1     1     7     1     1     7     1     6     7
## 2 1145       6     1     4     1     3     1     1     3     7     4     3     7
## 3 4808       4     3     3     3     2     5     3     2     6     4     4     4
## 4 5884       2     1     6     1     1     3     2     1     5     2     2     4
## 5 7015       2     1     7     3     2     5     2     2     6     2     2     6
## 6 1520       3     2     5     2     2     2     3     3     4     2     3     4
## # i 52 more variables: `13` <dbl>, `14` <dbl>, `15` <dbl>, `16` <dbl>,
## #   `17` <dbl>, `18` <dbl>, `19` <dbl>, `20` <dbl>, `21` <dbl>, `22` <dbl>,
## #   `23` <dbl>, `24` <dbl>, `25` <dbl>, `26` <dbl>, `27` <dbl>, `28` <dbl>,
## #   `29` <dbl>, `30` <dbl>, `31` <dbl>, `32` <dbl>, `33` <dbl>, `34` <dbl>,
## #   `35` <dbl>, `36` <dbl>, `37` <dbl>, `38` <dbl>, `39` <dbl>, `40` <dbl>,
## #   `41` <dbl>, `42` <dbl>, `43` <dbl>, `44` <dbl>, `45` <dbl>, `46` <dbl>,
## #   `47` <dbl>, `48` <dbl>, `49` <dbl>, `50` <dbl>, `51` <dbl>, `52` <dbl>, ...
```

```
fairness2_centered <- fairness2_filtered %>%
  mutate(across(c(2:65), ~ case_when(
    . == 1 ~ 3,
    . == 2 ~ 2,
    . == 3 ~ 1,
    . == 4 ~ 0,
    . == 5 ~ -1,
    . == 6 ~  -2,
    . == 7 ~ -3)))
head(fairness2_centered)
```

```
## # A tibble: 6 x 65
##   sjCode   `1`   `2`   `3`   `4`   `5`   `6`   `7`   `8`   `9`  `10`  `11`  `12`
##   <fct>  <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 7735       2     3    -2     3     3    -3     3     3    -3     3    -2    -3
## 2 1145      -2     3     0     3     1     3     3     1    -3     0     1    -3
## 3 4808       0     1     1     1     2    -1     1     2    -2     0     0     0
## 4 5884       2     3    -2     3     3     1     2     3    -1     2     2     0
## 5 7015       2     3    -3     1     2    -1     2     2    -2     2     2    -2
## 6 1520       1     2    -1     2     2     2     1     1     0     2     1     0
## # i 52 more variables: `13` <dbl>, `14` <dbl>, `15` <dbl>, `16` <dbl>,
## #   `17` <dbl>, `18` <dbl>, `19` <dbl>, `20` <dbl>, `21` <dbl>, `22` <dbl>,
## #   `23` <dbl>, `24` <dbl>, `25` <dbl>, `26` <dbl>, `27` <dbl>, `28` <dbl>,
## #   `29` <dbl>, `30` <dbl>, `31` <dbl>, `32` <dbl>, `33` <dbl>, `34` <dbl>,
```

```
## #   `35` <dbl>, `36` <dbl>, `37` <dbl>, `38` <dbl>, `39` <dbl>, `40` <dbl>,
## #   `41` <dbl>, `42` <dbl>, `43` <dbl>, `44` <dbl>, `45` <dbl>, `46` <dbl>,
## #   `47` <dbl>, `48` <dbl>, `49` <dbl>, `50` <dbl>, `51` <dbl>, `52` <dbl>, ...
```

```
fairness2_centered_pivot <- fairness2_centered %>%
  pivot_longer(!sjCode, names_to = "Trial", values_to = "F_rating") %>%
  arrange(sjCode) %>%
  mutate_at("Trial", as.numeric)
head(fairness2_centered_pivot)
```

```
## # A tibble: 6 x 3
##   sjCode Trial F_rating
##   <fct>  <dbl>    <dbl>
## 1 1145       1       -2
## 2 1145       2        3
## 3 1145       3        0
## 4 1145       4        3
## 5 1145       5        1
## 6 1145       6        3
```

```
fairness2_filtered_pivot <- fairness2_filtered %>%
  pivot_longer(!sjCode, names_to = "Trial", values_to = "F_filt_rating") %>%
  arrange(sjCode) %>%
  mutate_at("Trial", as.numeric)
head(fairness2_filtered_pivot)
```

```
## # A tibble: 6 x 3
##   sjCode Trial F_filt_rating
##   <fct>  <dbl>         <dbl>
## 1 1145       1             6
## 2 1145       2             1
## 3 1145       3             4
## 4 1145       4             1
## 5 1145       5             3
## 6 1145       6             1
```

```
table(fairness2_filtered_pivot$F_filt_rating)
```

```
##
##   1   2   3   4   5   6   7
## 172 127 122 132 183 108 116
```

```
table(fairness2_centered_pivot$F_rating)
```

```
##
##  -3  -2  -1   0   1   2   3
## 116 108 183 132 122 127 172
```

save the questionnaire data to new csv file

```r
write_csv(fairness2_centered_pivot, file = paste0(getwd(),"/Qualtrics/fairness_2_EMG4_final.csv"))
```

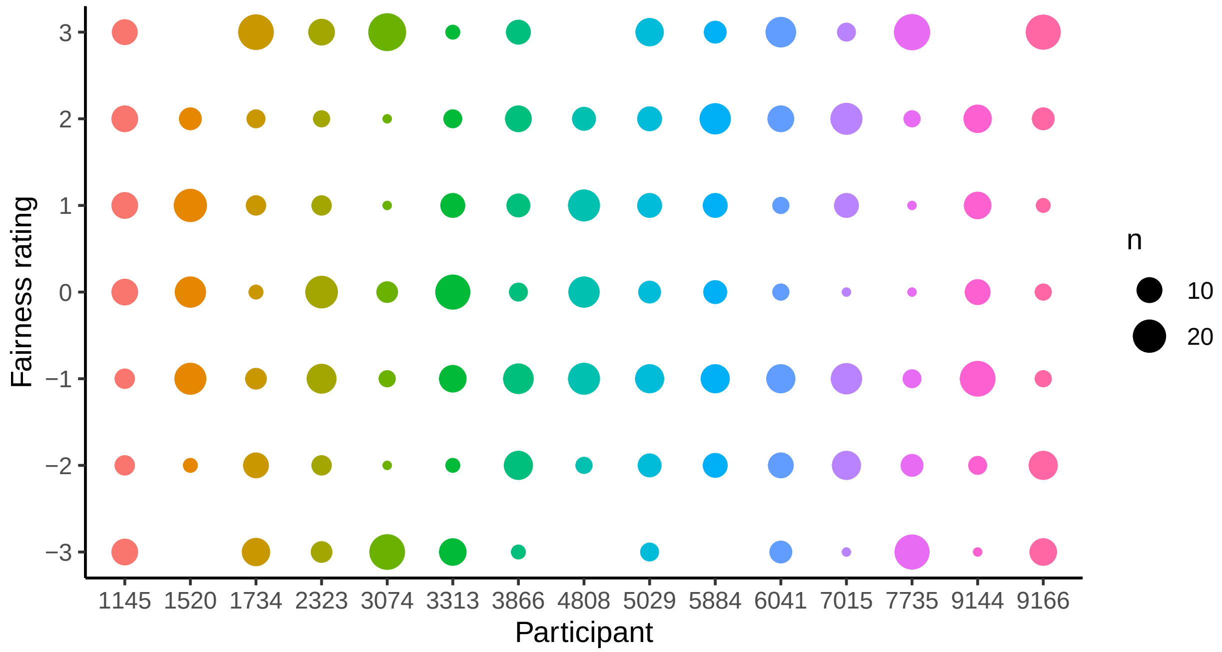

Figure 2: Variation in fairness ratings within participants (List 2)

Fairness List 3

```r
fairness3_filtered <- read_csv(paste0(getwd(),"/Qualtrics/EMG4_Vragenlijst_1.3_final.csv"), show_col_typ
  slice(-c(1, 2)) %>%
  select(as.character(0:64))%>%
  mutate_at('0', as.factor) %>%
  mutate_at(as.character(1:64), as.numeric) %>%
  rename(sjCode = '0')

head(fairness3_filtered)
```

```
## # A tibble: 6 x 65
##   sjCode   `1`   `2`   `3`   `4`   `5`   `6`   `7`   `8`   `9`  `10`  `11`  `12`
##   <fct>  <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 7417       4     4     4     4     3     3     4     5     6     5     4     5
## 2 9600       1     1     1     7     7     4     4     2     4     3     7     7
## 3 8800       2     2     1     7     7     1     6     1     6     2     7     6
## 4 8381       3     3     2     4     4     5     5     3     5     4     6     6
## 5 4806       4     4     4     4     4     4     4     4     4     4     4     4
## 6 3555       3     2     2     7     6     1     4     2     5     3     6     6
## # i 52 more variables: `13` <dbl>, `14` <dbl>, `15` <dbl>, `16` <dbl>,
## #   `17` <dbl>, `18` <dbl>, `19` <dbl>, `20` <dbl>, `21` <dbl>, `22` <dbl>,
## #   `23` <dbl>, `24` <dbl>, `25` <dbl>, `26` <dbl>, `27` <dbl>, `28` <dbl>,
```

```
## #   `29` <dbl>, `30` <dbl>, `31` <dbl>, `32` <dbl>, `33` <dbl>, `34` <dbl>,
## #   `35` <dbl>, `36` <dbl>, `37` <dbl>, `38` <dbl>, `39` <dbl>, `40` <dbl>,
## #   `41` <dbl>, `42` <dbl>, `43` <dbl>, `44` <dbl>, `45` <dbl>, `46` <dbl>,
## #   `47` <dbl>, `48` <dbl>, `49` <dbl>, `50` <dbl>, `51` <dbl>, `52` <dbl>, ...
```

```r
fairness3_centered <- fairness3_filtered %>%
  mutate(across(c(2:65), ~ case_when(
    . == 1 ~ 3,
    . == 2 ~ 2,
    . == 3 ~ 1,
    . == 4 ~ 0,
    . == 5 ~ -1,
    . == 6 ~ -2,
    . == 7 ~ -3)))
head(fairness3_centered)
```

```
## # A tibble: 6 x 65
##   sjCode   `1`   `2`   `3`   `4`   `5`   `6`   `7`   `8`   `9`  `10`  `11`  `12`
##   <fct>  <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 7417       0     0     0     0     1     1     0    -1    -2    -1     0    -1
## 2 9600       3     3     3    -3    -3     0     0     2     0     1    -3    -3
## 3 8800       2     2     3    -3    -3     3    -2     3    -2     2    -3    -2
## 4 8381       1     1     2     0     0    -1    -1     1    -1     0    -2    -2
## 5 4806       0     0     0     0     0     0     0     0     0     0     0     0
## 6 3555       1     2     2    -3    -2     3     0     2    -1     1    -2    -2
## # i 52 more variables: `13` <dbl>, `14` <dbl>, `15` <dbl>, `16` <dbl>,
## #   `17` <dbl>, `18` <dbl>, `19` <dbl>, `20` <dbl>, `21` <dbl>, `22` <dbl>,
## #   `23` <dbl>, `24` <dbl>, `25` <dbl>, `26` <dbl>, `27` <dbl>, `28` <dbl>,
## #   `29` <dbl>, `30` <dbl>, `31` <dbl>, `32` <dbl>, `33` <dbl>, `34` <dbl>,
## #   `35` <dbl>, `36` <dbl>, `37` <dbl>, `38` <dbl>, `39` <dbl>, `40` <dbl>,
## #   `41` <dbl>, `42` <dbl>, `43` <dbl>, `44` <dbl>, `45` <dbl>, `46` <dbl>,
## #   `47` <dbl>, `48` <dbl>, `49` <dbl>, `50` <dbl>, `51` <dbl>, `52` <dbl>, ...
```

```r
# note: 4806 change all ratings to NA because rated everything neutral,
# for 7417, 8381, 8800, and 9600 the ratings from 43 and up should be changed to NA
# (wrong story continuance was used)

fairness3_centered_pivot_old <- fairness3_centered %>%
  pivot_longer(!sjCode, names_to = "Trial", values_to = "F_rating") %>%
  arrange(sjCode) %>%
  mutate_at("Trial", as.numeric)

fairness3_centered_pivot <- fairness3_centered %>%
  pivot_longer(!sjCode, names_to = "Trial", values_to = "F_rating") %>%
  arrange(sjCode) %>%
  mutate_at("Trial", as.numeric)  %>%
  mutate(F_rating = case_when(sjCode == 4806 ~ as.numeric(NA),
                              sjCode == 7417 & Trial >= 43 ~ as.numeric(NA),
                              sjCode == 8381 & Trial >= 43 ~ as.numeric(NA),
                              sjCode == 8800 & Trial >= 43 ~ as.numeric(NA),
                              sjCode == 9600 & Trial >= 43 ~ as.numeric(NA),
                              TRUE ~ F_rating))
head(fairness3_centered_pivot)
```

```
## # A tibble: 6 x 3
##   sjCode Trial F_rating
##   <fct>  <dbl>    <dbl>
## 1 1055       1        1
## 2 1055       2        2
## 3 1055       3        2
## 4 1055       4       -3
## 5 1055       5       -3
## 6 1055       6        3
```

```
fairness3_filtered_pivot_old <- fairness3_filtered %>%
  pivot_longer(!sjCode, names_to = "Trial", values_to = "F_filt_rating") %>%
  arrange(sjCode) %>%
  mutate_at("Trial", as.numeric)

fairness3_filtered_pivot <- fairness3_filtered %>%
  pivot_longer(!sjCode, names_to = "Trial", values_to = "F_filt_rating") %>%
  arrange(sjCode) %>%
  mutate_at("Trial", as.numeric)%>%
  mutate(F_filt_rating = case_when(sjCode == 4806 ~ as.numeric(NA),
                                   sjCode == 7417 & Trial >= 43 ~ as.numeric(NA),
                                   sjCode == 8381 & Trial >= 43 ~ as.numeric(NA),
                                   sjCode == 8800 & Trial >= 43 ~ as.numeric(NA),
                                   sjCode == 9600 & Trial >= 43 ~ as.numeric(NA),
                                   TRUE ~ F_filt_rating))
head(fairness3_filtered_pivot)
```

```
## # A tibble: 6 x 3
##   sjCode Trial F_filt_rating
##   <fct>  <dbl>         <dbl>
## 1 1055       1             3
## 2 1055       2             2
## 3 1055       3             2
## 4 1055       4             7
## 5 1055       5             7
## 6 1055       6             1
```

```
table(fairness3_filtered_pivot_old$F_filt_rating)
```

```
##
##   1   2   3   4   5   6   7
## 192 158  96 152 136 145 137
```

```
table(fairness3_centered_pivot_old$F_rating)
```

```
##
##  -3  -2  -1   0   1   2   3
## 137 145 136 152  96 158 192
```

```
table(fairness3_filtered_pivot$F_filt_rating)
```

```
##
##   1   2   3   4   5   6   7
## 175 150  85  77 119 132 134
```

```
table(fairness3_centered_pivot$F_rating)
```

```
##
##  -3  -2  -1   0   1   2   3
## 134 132 119  77  85 150 175
```

save the questionnaire data to new csv file

```
write_csv(fairness3_centered_pivot, file = paste0(getwd(),"/Qualtrics/fairness_3_EMG4_final.csv"))
```

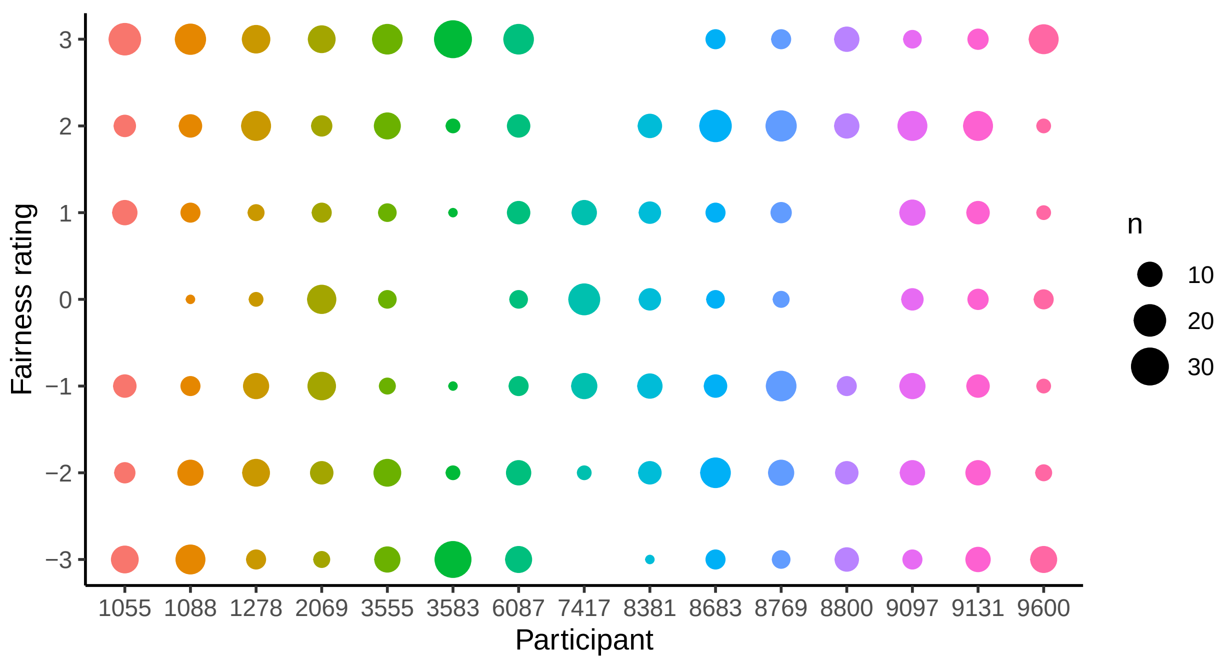

Figure 3: Variation in fairness ratings within participants (List 3)

Fairness List 4

```
fairness4_filtered <- read_csv(paste0(getwd(),"/Qualtrics/EMG4_Vragenlijst_1.4_final.csv")
                               , show_col_types = FALSE) %>%
  slice(-c(1, 2)) %>%
  select(as.character(0:64))%>%
  mutate_at('0', as.factor) %>%
  mutate_at(as.character(1:64), as.numeric) %>%
  rename(sjCode = '0')

head(fairness4_filtered)
```

```
## # A tibble: 6 x 65
##   sjCode   `1`   `2`   `3`   `4`   `5`   `6`   `7`   `8`   `9`  `10`  `11`  `12`
##   <fct>  <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 9080       6     5     1     4     3     1     6     5     1     3     4     2
## 2 5959       2     7     1     3     3     1     5     3     3     3     3     5
## 3 2020       3     5     1     2     2     5     6     5     1     2     2     4
## 4 7716       4     1     1     6     1     5     7     1     1     4     4     1
## 5 8320       5     6     2     3     6     5     5     3     3     6     6     1
## 6 3880       4     6     2     5     7     3     6     5     2     6     7     2
## # i 52 more variables: `13` <dbl>, `14` <dbl>, `15` <dbl>, `16` <dbl>,
## #   `17` <dbl>, `18` <dbl>, `19` <dbl>, `20` <dbl>, `21` <dbl>, `22` <dbl>,
## #   `23` <dbl>, `24` <dbl>, `25` <dbl>, `26` <dbl>, `27` <dbl>, `28` <dbl>,
## #   `29` <dbl>, `30` <dbl>, `31` <dbl>, `32` <dbl>, `33` <dbl>, `34` <dbl>,
## #   `35` <dbl>, `36` <dbl>, `37` <dbl>, `38` <dbl>, `39` <dbl>, `40` <dbl>,
## #   `41` <dbl>, `42` <dbl>, `43` <dbl>, `44` <dbl>, `45` <dbl>, `46` <dbl>,
## #   `47` <dbl>, `48` <dbl>, `49` <dbl>, `50` <dbl>, `51` <dbl>, `52` <dbl>, ...
```

```
fairness4_centered <- fairness4_filtered %>%
  mutate(across(c(2:65), ~ case_when(
    . == 1 ~ 3,
    . == 2 ~ 2,
    . == 3 ~ 1,
    . == 4 ~ 0,
    . == 5 ~ -1,
    . == 6 ~ -2,
    . == 7 ~ -3)))
head(fairness4_centered)
```

```
## # A tibble: 6 x 65
##   sjCode   `1`   `2`   `3`   `4`   `5`   `6`   `7`   `8`   `9`  `10`  `11`  `12`
##   <fct>  <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 9080      -2    -1     3     0     1     3    -2    -1     3     1     0     2
## 2 5959       2    -3     3     1     1     3    -1     1     1     1     1    -1
## 3 2020       1    -1     3     2     2    -1    -2    -1     3     2     2     0
## 4 7716       0     3     3    -2     3    -1    -3     3     3     0     0     3
## 5 8320      -1    -2     2     1    -2    -1    -1     1     1    -2    -2     3
## 6 3880       0    -2     2    -1    -3     1    -2    -1     2    -2    -3     2
## # i 52 more variables: `13` <dbl>, `14` <dbl>, `15` <dbl>, `16` <dbl>,
## #   `17` <dbl>, `18` <dbl>, `19` <dbl>, `20` <dbl>, `21` <dbl>, `22` <dbl>,
## #   `23` <dbl>, `24` <dbl>, `25` <dbl>, `26` <dbl>, `27` <dbl>, `28` <dbl>,
## #   `29` <dbl>, `30` <dbl>, `31` <dbl>, `32` <dbl>, `33` <dbl>, `34` <dbl>,
## #   `35` <dbl>, `36` <dbl>, `37` <dbl>, `38` <dbl>, `39` <dbl>, `40` <dbl>,
## #   `41` <dbl>, `42` <dbl>, `43` <dbl>, `44` <dbl>, `45` <dbl>, `46` <dbl>,
## #   `47` <dbl>, `48` <dbl>, `49` <dbl>, `50` <dbl>, `51` <dbl>, `52` <dbl>, ...
```

```
# note: for 2020, 5959, and 9080 the rating for trial 13 should be changed to NA
# (morality manipulation was missing)
fairness4_centered_pivot <- fairness4_centered %>%
  pivot_longer(!sjCode, names_to = "Trial", values_to = "F_rating") %>%
  arrange(sjCode) %>%
  mutate_at("Trial", as.numeric) %>%
  mutate(F_rating = case_when(sjCode == 2020 & Trial ==13 ~ as.numeric(NA),
                              sjCode == 5959 & Trial ==13 ~ as.numeric(NA),
```

```r
                              sjCode == 9080 & Trial ==13 ~ as.numeric(NA),
                              TRUE ~ F_rating))
head(fairness4_centered_pivot)
```

```
## # A tibble: 6 x 3
##   sjCode Trial F_rating
##   <fct>  <dbl>    <dbl>
## 1 1124       1       -2
## 2 1124       2       -1
## 3 1124       3        3
## 4 1124       4        0
## 5 1124       5       -3
## 6 1124       6        3
```

```r
fairness4_filtered_pivot <- fairness4_filtered %>%
  pivot_longer(!sjCode, names_to = "Trial", values_to = "F_filt_rating") %>%
  arrange(sjCode) %>%
  mutate_at("Trial", as.numeric) %>%
  mutate(F_filt_rating = case_when(sjCode == 2020 & Trial ==13 ~ as.numeric(NA),
                              sjCode == 5959 & Trial ==13 ~ as.numeric(NA),
                              sjCode == 9080 & Trial ==13 ~ as.numeric(NA),
                              TRUE ~ F_filt_rating))
head(fairness4_filtered_pivot)
```

```
## # A tibble: 6 x 3
##   sjCode Trial F_filt_rating
##   <fct>  <dbl>         <dbl>
## 1 1124       1             6
## 2 1124       2             5
## 3 1124       3             1
## 4 1124       4             4
## 5 1124       5             7
## 6 1124       6             1
```

```r
table(fairness4_filtered_pivot$F_filt_rating)
```

```
##
##   1   2   3   4   5   6   7
## 169 142 122 120 147 132  61
```

```r
table(fairness4_centered_pivot$F_rating)
```

```
##
##  -3  -2  -1   0   1   2   3
##  61 132 147 120 122 142 169
```

save the questionnaire data to new csv file

```r
write_csv(fairness4_centered_pivot, file = paste0(getwd(),"/Qualtrics/fairness_4_EMG4_final.csv"))
```

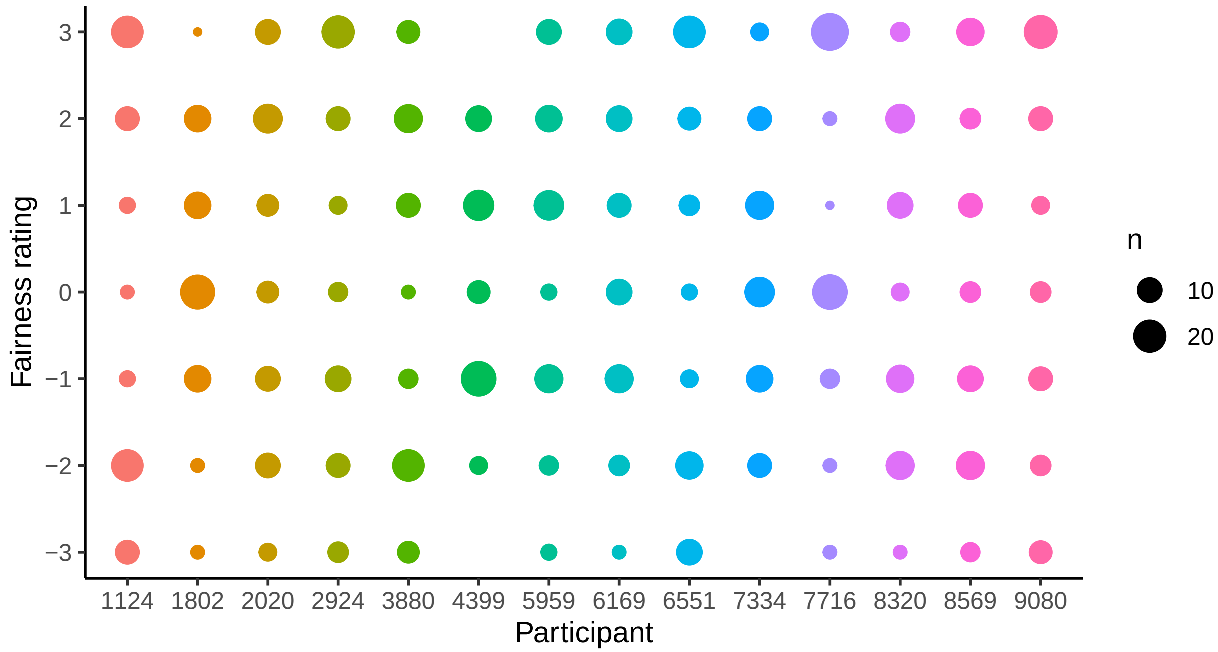

Figure 4: Variation in fairness ratings within participants (List 4)

Combine all fairness data lists

```r
fairness1_centered_pivot<-read_csv(paste0(getwd(), "/Qualtrics/fairness_1_EMG4_final.csv"))
fairness2_centered_pivot<-read_csv(paste0(getwd(), "/Qualtrics/fairness_2_EMG4_final.csv"))
fairness3_centered_pivot<-read_csv(paste0(getwd(), "/Qualtrics/fairness_3_EMG4_final.csv"))
fairness4_centered_pivot<-read_csv(paste0(getwd(), "/Qualtrics/fairness_4_EMG4_final.csv"))


# put all fairness data frames into list
fairness_all <- bind_rows(fairness1_centered_pivot, fairness2_centered_pivot, fairness3_centered_pivot,

fairness_all_sorted <- bind_rows(fairness1_centered_pivot, fairness2_centered_pivot, fairness3_centered_
  arrange(sjCode, Trial)

head(fairness_all)
```

```
## # A tibble: 6 x 3
##   sjCode Trial F_rating
##    <dbl> <dbl>    <dbl>
## 1   1258     1       -2
## 2   1258     2       -1
## 3   1258     3       -2
## 4   1258     4        2
```

```
## 5   1258     5        2
## 6   1258     6       -3
```

```r
# put all fairness data frames into list
fairness_all_filtered <- bind_rows(fairness1_filtered_pivot, fairness2_filtered_pivot, fairness3_filtere
  arrange(sjCode, Trial)

head(fairness_all_filtered)
```

```
## # A tibble: 6 x 3
##   sjCode Trial F_filt_rating
##   <fct>  <dbl>         <dbl>
## 1 1258       1             6
## 2 1258       2             5
## 3 1258       3             6
## 4 1258       4             2
## 5 1258       5             2
## 6 1258       6             7
```

```r
table(fairness_all_filtered$F_filt_rating)
```

```
##
##   1   2   3   4   5   6   7
## 694 583 434 442 599 525 408
```

```r
table(fairness_all$F_rating)
```

```
##
##  -3  -2  -1   0   1   2   3
## 408 525 599 442 434 583 694
```

save the fairness data to new csv file

```r
write_csv(fairness_all, file = paste0(getwd(),"/Qualtrics/fairness_all_EMG4_final.csv"))
```